

Entropie

Relative Entropie von endlichen Quantenzuständen

Moritz Lanz

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Abstract

This is the abstract.

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1 Section 1

1.1 Erstmal die Basics

Weil ich alles vergesse haha.

Definition 1.1. Sei $\mathcal{M} \subset \mathcal{B}(H)$ eine von Neumann Algebra. Das Predual von $\mathcal{M}$ (geschrieben $\mathcal{M}_*$) ist der Raum aller σ -schwachen stetigen linearen Funktionale auf $\mathcal{M}$.

Definition 1.2. Seien $(v_n)_{n \in \mathbb{N}}$ und $(w_n)_{n \in \mathbb{N}}$ zwei Folgen in $\mathcal{H}$, so dass gilt:

$$\sum_n \|v_n\|^2 < \infty, \quad \sum_n \|w_n\|^2 < \infty. \quad (1)$$

Die σ -schwache Topologie auf $\mathcal{B}(\mathcal{H})$ ist definiert als die lokal konvexe Topologie erzeugt durch folgende Seminormen:

$$p_{(v_n)_n, (w_n)_n}(A) := \left| \sum_n \langle v_n, Aw_n \rangle \right|, \quad A \in \mathcal{B}(\mathcal{H}). \quad (2)$$

Definition 1.3. Sei $\mathcal{A}$ eine C^* -Algebra. Der positive Kegel von $\mathcal{A}$ ist definiert als die Menge aller positiven Elemente aus $\mathcal{A}$. Wir schreiben $\mathcal{A}_+$.

1.2 Some Definitions

Question 1.4. Definieren hier die f-Divergenz allgemein für positive lineare Funktionale über von Neumann Algebren. Haben wir dadurch später Vorteile, oder willst du das einfach erstmal so allgemein wie möglich machen? Für die Physik und viele Resultate werden wir schätze ich mal Zustände verwenden.

Definition 1.5. Let $\varphi, \psi : \mathcal{M} \rightarrow \mathbb{C}$ positive linear functionals. We call φ absolutely continuous with respect to ψ and write $\varphi \ll \psi$ if:

$$\psi(A^*A) = 0 \implies \varphi(A^*A) = 0 \quad \forall A \in \mathcal{M}. \quad (3)$$

We call φ and ψ mutually absolutely continuous and write $\varphi \ll\!\!\gg \psi$ if:

$$\psi(A^*A) = 0 \iff \varphi(A^*A) = 0. \quad (4)$$

For normal linear functionals we get additional information from mutual continuity. Therefore we first revise some properties.

Wir wollen später annehmen, dass zwei positive lineare Funktionale absolut stetig zueinander sind $\varphi \ll\!\!\gg \psi$, bzw. mindestens ein Funktional absolut stetig bzgl eines anderen ist $\varphi \ll \psi$. Um mehr information aus dieser Eigenschaft ziehen zu können fordern wir, dass die Funktionale im Folgenden normal sind. Dadurch existiert zu jedem positiv normalen linearen Funktional ein eindeutiger Dichteoperator.

Theorem 1.6. Let $\mathcal{M}$ be a von Neumann algebra and $\omega \in S(\mathcal{M})$. Then, ω is a normal state if and only if there exists a density operator $\rho \in \mathcal{M}$ such that $\omega(A) = \text{Tr}(\rho A)$ for all $A \in \mathcal{M}$.

Definition 1.7. Let $\mathcal{M}$ be an arbitrary von Neumann algebra. A density matrix/operator $\rho \in \mathcal{M}$ is a positive selfadjoint

Definition 1.8. Let $\varphi, \psi : \mathcal{M} \rightarrow \mathbb{C}$ be positive linear functionals over a von Neumann algebra $\mathcal{M}$ such that $\varphi \ll\!\!\gg \psi$ and $f : (0, \infty) \rightarrow \mathbb{R}$ be a convex function with $f(1) = 0$ (brauche ich das hier???). The f -divergence of φ and ψ is given by

$$\begin{aligned} D_f(\varphi|\psi) &= \int_1^\infty f''(\gamma) E_\gamma(\varphi|\psi) + \gamma^{-3} f''(\gamma^{-1}) E_\gamma(\psi|\varphi) d\gamma \\ &= \int_1^\infty f''(\gamma) E_\gamma(\varphi|\psi) d\gamma + \int_0^1 \gamma f''(\gamma) E_{\frac{1}{\gamma}}(\psi|\varphi) d\gamma. \end{aligned} \quad (5)$$

Definition 1.9. Let φ and ψ be as above. (sicher? oder doch Zustände?) The hockeystick divergence $E_\gamma(\varphi|\psi)$ is defined as follows:

$$E_\gamma(\varphi|\psi) := (\psi - \gamma\varphi)_+(\mathbb{1}) \quad (6)$$

Some facts about the hockeystick divergence.

Proposition 1.10. Let φ_1, φ_2 be positive linear functionals over a von Neumann algebra, and p_1, p_2 are (pos. lin. functionals or states?)

$$\lim_{t \rightarrow \infty} E_t(\varphi_1|\varphi_2) = \varphi_1(p_2^\perp). \quad (7)$$

$$\lim_{t \rightarrow \infty} E_t(\varphi_1, \varphi_2) = 0 \iff p_1 \leq p_2. \quad (8)$$

1.3 max-relative entropy

Definition 1.11. Let $\varphi, \psi : \mathcal{M} \rightarrow \mathbb{C}$ be two positive linear functionals. The *max-relative entropy* of φ with respect to ψ is defined as:

$$D_{\max}(\varphi|\psi) := \inf\{\lambda \in \mathbb{R} \mid \varphi \leq e^\lambda \psi\}. \quad (9)$$

Remark. In mathematical physics e^λ is preferred while in quantum information theory 2^λ is used because of bits (information unit).

Remark. If φ and ψ are states with $\varphi \leq \psi$, then $\varphi = \psi$ holds. Therefore, we define the positive linear functional $\omega = \psi - \varphi$. By the normalized property of φ and ψ we get:

$$\omega(\mathbb{1}) = \psi(\mathbb{1}) - \varphi(\mathbb{1}) = 1 - 1 = 0. \quad (10)$$

Hence, $\|\omega\| = \omega(\mathbb{1}) = 0$. By the positive definiteness of the norm we get that ω is the zero functional, thus $\varphi = \psi$.

1.4 First Inequalities

In [SV16] there are a lot of inequalities for f-divergences in the context of probability measures.

We want to generalize them for states over arbitrary von Neumann algebras (even vNA of type III).

We start with inequality (84) from [SV16].

Question 1.12. Wann ist $D_f(\varphi|\psi)$ endlich? Evtl wenn ψ normal ist?

Proposition 1.13. Let $f, g : (0, \infty) \rightarrow \mathbb{R}$ convex functions with $f(1) = g(1)$, $f'(1) = g'(1)$ and $f(t) \leq g(t)$ for all $t \in (0, \infty)$. Also let φ, ψ be positive linear functionals over a von Neumann algebra such that $\varphi \ll \psi$ ($D_f(\varphi|\psi), D_g(\varphi|\psi)$ are finite)

$$D_f(\varphi|\psi) \leq D_g(\varphi|\psi). \quad (11)$$

Remark. Alexandrov's theorem states that every convex function defined on an open domain is twice differentiable almost everywhere.

Remark. By assumption $f'(1) = g'(1)$ indirectly require, that f and g are differentiable at $x = 1$. Therefore, we cannot put in e.g. the *total variational distance* (see [SV16] eq. (55)).

Proof. We begin by reformulating the f-divergence given in definition 1.8.

$$\begin{aligned}
D_f(\varphi|\psi) &= \int_1^\infty f''(\gamma) E_\gamma(\varphi|\psi) + \gamma^{-3} f''(\gamma^{-1}) E_\gamma(\psi|\varphi) d\gamma \\
&= \int_1^\infty f''(\gamma) E_\gamma(\varphi|\psi) d\gamma + \int_0^1 \gamma f''(\gamma) E_{\frac{1}{\gamma}}(\psi|\varphi) d\gamma \\
&\stackrel{(a)}{=} \int_1^\infty f''(\gamma) E_\gamma(\varphi|\psi) d\gamma + \int_0^1 \gamma f''(\gamma) \left(\frac{1}{\gamma} E_\gamma(\varphi|\psi) + \psi(\mathbb{1}) - \frac{1}{\gamma} \varphi(\mathbb{1}) \right) d\gamma \\
&= \int_1^\infty f''(\gamma) E_\gamma(\varphi|\psi) d\gamma + \int_0^1 f''(\gamma) E_\gamma(\varphi|\psi) d\gamma + \int_0^1 f''(\gamma) (\gamma \psi(\mathbb{1}) - \varphi(\mathbb{1})) d\gamma \\
&= \int_0^\infty f''(\gamma) E_\gamma(\varphi|\psi) d\gamma + \psi(\mathbb{1}) \int_0^1 \gamma f''(\gamma) d\gamma - \varphi(\mathbb{1}) \int_0^1 f''(\gamma) d\gamma \\
&\stackrel{(b)}{=} [f'(\gamma) E_\gamma(\varphi|\psi)]_0^\infty - \int_0^\infty f'(\gamma) E'_\gamma(\varphi|\psi) d\gamma + \psi(\mathbb{1}) \left([\gamma f'(\gamma)]_0^1 - \int_0^1 f'(\gamma) d\gamma \right) - \varphi(\mathbb{1}) [f'(\gamma)]_0^1 \\
&\stackrel{(c)}{=} [f'(\gamma) E_\gamma(\varphi|\psi)]_0^\infty - \left([f(\gamma) E'_\gamma(\varphi|\psi)]_0^\infty - \int_0^\infty f(\gamma) E''_\gamma(\varphi|\psi) d\gamma \right) \\
&\quad + \psi(\mathbb{1}) ([\gamma f'(\gamma)]_0^1 - [f(\gamma)]_0^1) - \varphi(\mathbb{1}) [f'(\gamma)]_0^1 \\
&= \lim_{\gamma \rightarrow \infty} f'(\gamma) E_\gamma(\varphi|\psi) - \lim_{\gamma \rightarrow 0} f'(\gamma) E_\gamma(\varphi|\psi) - \left(\lim_{\gamma \rightarrow \infty} f(\gamma) E'_\gamma(\varphi|\psi) - \lim_{\gamma \rightarrow 0} f(\gamma) E'_\gamma(\varphi|\psi) \right) \\
&\quad + \int_0^\infty f(\gamma) E''_\gamma(\varphi|\psi) d\gamma + \psi(\mathbb{1}) \left(f'(1) - \lim_{\gamma \rightarrow 0} \gamma f'(\gamma) - f(1) + \lim_{\gamma \rightarrow 0} f(\gamma) \right) \\
&\quad - \varphi(\mathbb{1}) \left(f'(1) - \lim_{\gamma \rightarrow 0} f'(\gamma) \right) \\
&= \lim_{\gamma \rightarrow \infty} (f'(\gamma) E_\gamma(\varphi|\psi) - f(\gamma) E'_\gamma(\varphi|\psi)) + \int_0^\infty f(\gamma) E''_\gamma(\varphi|\psi) d\gamma \\
&\quad + \lim_{\gamma \rightarrow 0} f(\gamma) (E'_\gamma(\varphi|\psi) + \psi(\mathbb{1})) + \lim_{\gamma \rightarrow 0} f'(\gamma) (\varphi(\mathbb{1}) - \gamma \psi(\mathbb{1}) - E_\gamma(\varphi|\psi)) \\
&\quad + \underbrace{\psi(\mathbb{1}) f'(1) - \psi(\mathbb{1}) f(1) - \varphi(\mathbb{1}) f'(1)}_{=:K}
\end{aligned} \quad (12)$$

(a) state exchange, (b), (c) partial integration K is a constant here, fällt weg bei $D_g - D_f$, da $f(1) = g(1)$ und $f'(1) = g'(1)$.

(a)

Next we want to show under which conditions $D_g(\varphi|\psi) \leq D_f(\varphi|\psi)$ holds.

$$\begin{aligned}
D_g(\varphi|\psi) - D_f(\varphi|\psi) &= \int_0^\infty \underbrace{(g(\gamma) - f(\gamma))}_{\geq 0} \underbrace{E''_\gamma(\varphi|\psi)}_{\geq 0} d\gamma + \lim_{\gamma \rightarrow 0} \underbrace{(g(\gamma) - f(\gamma))}_{\geq 0} \underbrace{(E'_\gamma(\varphi|\psi) + \psi(\mathbb{1}))}_{\geq 0 \text{ (a)}} \\
&+ \lim_{\gamma \rightarrow \infty} ((g'(\gamma) - f'(\gamma))E_\gamma(\varphi|\psi) - (g(\gamma) - f(\gamma))E'_\gamma(\varphi|\psi)) \\
&+ \lim_{\gamma \rightarrow 0} (g'(\gamma) - f'(\gamma)) \underbrace{(\varphi(\mathbb{1}) - \gamma\psi(\mathbb{1}) - E_\gamma(\varphi|\psi))}_{\leq 0 \text{ (b)}}
\end{aligned} \tag{13}$$

(a) follow from lipschitz-continuity of E_γ ; (b) follows from strict positivity of E_γ .

□

2 Section 2

2.1 Subsection A

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References

- [SV16] Igal Sason and Sergio Verdu. “ f -Divergence Inequalities”. In: *IEEE Transactions on Information Theory* 62.11 (Nov. 2016), pp. 5973–6006. ISSN: 1557-9654. DOI: 10.1109/tit.2016.2603151. URL: <http://dx.doi.org/10.1109/TIT.2016.2603151>.